



Accessly

Kraków without barriers

Your city. Your needs. Check before you go.

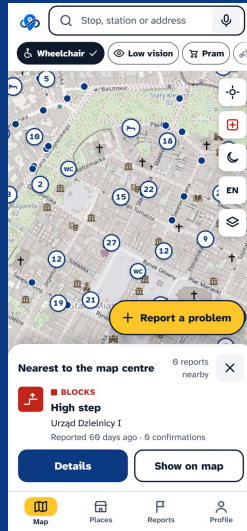
A guide to accessible places and a map of barriers in public space. Every piece of information carries its source, date and verification status.

Team Accessly · Team ID: ...
HackYeah 2026 · “Cracow without barriers” challenge

Demo: <https://accessly.example.pl>

Code: <https://github.com/HakerSII/accessly>

Video (3 min): <https://youtu.be/...>



“Accessible / not accessible” is not enough

Problem

- One label does not say whether I can get in with a wheelchair, whether the door is 90 cm wide, or whether there is a toilet.
- Information online has no source and no date: a broken lift turns “accessible” into a wasted trip.
- Missing information is shown as if it were confirmation, and the City has no capacity to maintain a database by hand.

Who it is for – scope of the prototype

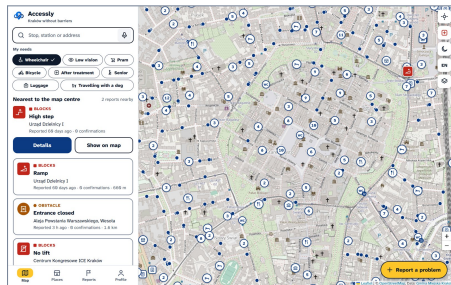
- **Primary group:** wheelchair users and parents with prams.
- The same data serves further profiles: Low vision, Senior, After treatment, Luggage, Dog, Bicycle.
- Residents and tourists (in Polish, English, German and Ukrainian) and **venue owners**, who get their own panel.

Our answer

Accessly answers “**will this place work for me?**” with concrete facts, each with a **source, a date and a verification status**. No data means no data, never accessibility.

A working prototype: a web app for phone and desktop

- **Catalogue of 6,545 places** (OpenStreetMap), 7 categories, **27 attributes**: steps, ramp, lift, doors ≥ 90 cm, toilet, changing table, hearing loop, places to rest. . .
- **Needs profile** instead of a question about disability: it filters, sorts, marks “Matches your needs” and says what is missing.
- **Place card**: every fact with its source, date and status; Up to date / Outdated / Correct / Add, a question to the owner, reviews, photos, a wheelchair route.
- **Barrier map** (a “Waze for accessibility”): 10 report types, blocks / obstacle severity, “still there” / “fixed” votes, automatic expiry.
- **Natural-language assistant** (Claude with a search tool over the database; keyword rules without a key): it shows only facts from the database.
- **11 open-data layers**, owner and admin panels, 4 languages, night mode.



“Wheelchair” profile: places, layers, reported barriers with severity and confirmations. The list next to the map is its text alternative.

6,545

places from OSM

11

data layers

503

tests, offline

Demo: Anna, a wheelchair user, looks for a café in the centre

Step	What to look at
1	Needs profile "Wheelchair" the map and lists show only barriers that matter for a wheelchair; ranking by match
2	Question: "a café in the centre, wheelchair, with a toilet" a ranking with reasons: what the place meets and what is <i>unknown</i>
3	Place card every fact has a source, a date and a status; missing data is named as such, with an "Add" button
4	Report "No lift" at a stop, severity "Blocks" square marker = blocks; "Still there" / "Fixed" votes; the outage expires after 14 days
5	Owner panel: confirm the information status "Confirmed by the owner"; Anna gets a notification

Edge cases shown in the demo: conflicting data → "Needs re-checking"; city data unavailable → snapshot with an "as of" date; missing information never counts as accessible.

Sample data is marked: demo accounts (anna@, kasia@, admin@accessly.test), test reports, a "Demo mode" label.

Meets 0 of your 3 needs

Does not meet: Step-free entrance, Accessible toilet

No data: Wide doors (at least 90 cm). If you know, add it below or ask the owner.

Step 3: the match summary – what the place meets, what it does not, what is unknown.

 **Step-free entrance** · important to you No

From an official source

Source: OpenStreetMap

02/10/2026

częściowo dostępne (wheelchair=limited)

Up to date

Outdated

Correct

One fact: value, source (OpenStreetMap), date (2 Oct 2026), status and the Up to date / Outdated / Correct buttons.

Data: sources, updates, trustworthiness

Source	What it provides	Updates and fallback when the source is down
OpenStreetMap (ODbL)	6,545 places with attributes from tags (wheelchair, toilets:wheelchair, changing_table...); lifts, kerbs, crossings with tactile paving, luggage storage, A&E and 24-hour pharmacies	Geofabrik extract → build_osm_*.py scripts → snapshot in the repository (as of 2 Oct 2026, date shown in the app); the import never overwrites values from owners and users
Kraków open data (ArcGIS, ot-wartedane.um.krakow.pl)	stops with platform kerb type (3,756), public toilets (50), disabled parking spaces (2,037), bike routes (2,633), bike stands (3,157), bike infrastructure (199)	fetches in pages of 2,000, 24-hour cache, refreshed in the background; when the service is silent: cache → repository snapshot with an “as of” date → fetch on demand; city attribution on the map
Venue owner	attribute changes, confirmations, photos, points inside the venue (lifts, toilets, entrances)	visible at once as “Confirmed by the owner”; claiming a profile is approved by a moderator
Community	Up to date / Outdated / Correct / Add, reviews, questions, photos, barrier reports and votes	corrections accepted by the owner or a moderator; 2 confirmations = “Confirmed by the community”; every change stays in the history

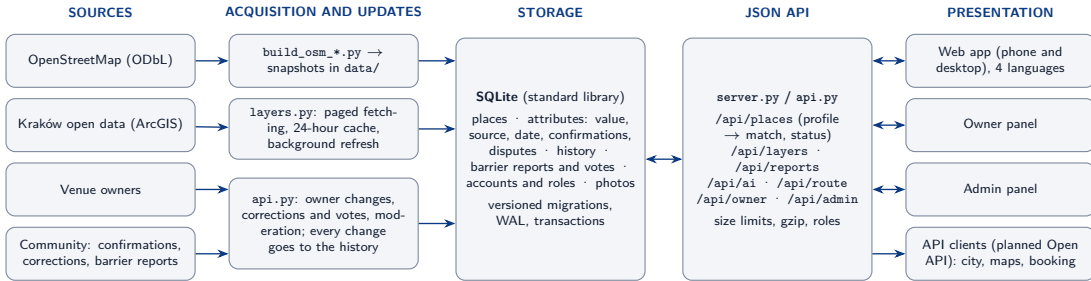
Status of every fact (computed on every read, never stored once):

Confirmed by the owner > Confirmed by the community (≥ 2) > From an official source > Not verified

More “outdated” votes than confirmations, or 365 days without confirmation → Needs re-checking.

Rules. Missing information never means “accessible”; conflicting data is flagged, not averaged; the AI assistant never adds facts. Barrier reports expire (permanent 365 days, lift outages 14, closed entrance 7); “still there” extends, 3 “fixed” votes close.

Architecture: data acquisition separated from presentation



Optional services, each with a fallback: Claude API (recommendations; keyword rules without a key) · OpenRouteService (wheelchair routes) · Nominatim (search from the browser, only on submit) · Rampa backend (barrier reports through the api.js adapter)

Zero dependencies. Python 3.7+ (standard library only), SQLite, Leaflet; 47 MB Docker image; Ansible + Caddy (HTTPS) on any VPS; 503 tests, no network.

New city. cities/<city>.json (name, bounds) + OSM extract + city layers (ArcGIS / WFS) in layers.py; ACCESSLY_CITY.

New source, category, attribute. An entry in the catalogue (LAYERS, CATEGORIES, ATTRIBUTES) + labels in 4 languages; consistency tests.

Digital accessibility: WCAG 2.2 AA as the goal

In the prototype (checked on the main scenario)

- **Keyboard:** every action is a button or a link; a “Skip to content” link; visible focus that survives the list refresh every 15 s; Esc closes panels.
- **Screen reader:** landmarks, headings, ARIA labels, short live announcements; the map’s description points to the list; the window title follows the view.
- **Map as text:** the Reports, Places and nearest-layer-items lists; severity by shape and label, not by colour alone.
- **Contrast and legibility:** colour tokens 4.5:1 (text) and 3:1 (controls) checked by a test; night mode; the Atkinson Hyperlegible typeface; 4 languages.
- **Forms and touch:** errors in `role="alert"` naming the field, “Next” is never disabled; targets ≥ 44 px; the layout works at 400 % zoom.

Still to do

- An audit with NVDA / VoiceOver and tests with wheelchair users and people with low vision.
- Keyboard access to map markers (Leaflet); an alternative to placing the pin by panning the map (2.5.7).
- Owner and admin panels to the same standard, and in English.
- An accessibility statement and a conformance report.

The “WCAG 2.2 A” and “WCAG 2.2 AA” iterations are covered by the `AccessibilityTests` suite.

Privacy, security and operations outside the City

User data and security

- **Without an account:** map, catalogue and barrier reports, anonymously; no IP addresses or device identifiers are stored.
- **The needs profile is a set of preferences** (steps, lift, quiet), never a diagnosis or disability information, exactly as the brief asks.
- **Passwordless account:** e-mail + one-time code (15 min, 5 attempts); codes and tokens stored as SHA-256 hashes; HttpOnly, SameSite, Secure cookie; one-click account deletion.
- **Server:** HTTPS (Caddy, Let's Encrypt), request limits (10 kB, photos 4 MB), user content always escaped (tested), roles checked on every request; privacy policy inside the app.

Launch and operations – proposal

- **Operator:** the Accessly team (company or foundation) or a community partner; the City maintains no database and grants no access to internal systems.
- **Cost (estimate):** one VPS in the EU at about 60–120 PLN (15–30 EUR) a month, a domain, transactional e-mail; the AI assistant is billed per request (fractions of a cent).
- **Updates** with one command (Ansible); city data refreshes itself every 24 h, OSM by script once a month; SQLite backups.
- **Reports and moderation** in the admin panel (queue, claims, corrections); portability: Docker on any infrastructure, open licences listed in the app.

Business model and commercialisation

Segment	Offer	Revenue
Residents and tourists	the app, needs profile, assistant, barrier map	free, always – they create the data
Venue owners (hotels, food & drink, culture, chains)	basic panel free; Pro plan : owner badge, statistics on “what guests look for and what is missing”, bulk facilities, an accessibility widget for the website and the booking system	subscription per venue or a package for chains
Event organisers, property managers	time-limited package: temporary entrances and routes, a barrier map for the event	fee per event / venue
Cities and regions	white-label deployment (configuration, OSM import, city layers); reports on gaps in data and facilities	deployment + annual maintenance
Maps, travel apps, booking systems	Open API with attribution: accessibility facts with source, date and status	API licence / request limits

Principles that protect the value of the data. We do not sell personal data; community content and OSM data stay open (ODbL). Owners pay for tools and visibility, not for a “better” status – the status always follows verification.

From prototype to service

What we have today

- A prototype for phone and desktop: 6,545 places, 27 attributes, 11 layers, barrier map, assistant, 4 languages.
- An owner panel and an admin panel (queue, statistics, data gaps).
- Source, date and status on every fact; missing data shown as missing.
- Checked WCAG 2.2 A/AA criteria on the main scenario; 503 tests, Docker, Ansible, HTTPS.

Plan for the next 3–6 months

- ① **Pilot** with a disability organisation, 2–3 hotels and cultural institutions; an accessibility audit with users.
- ② **Production sign-in** (transactional e-mail, optional OAuth), notifications, offline mode.
- ③ **AI for checking photos and text**: suggestions to confirm, never facts; the trust engine from Rampa.
- ④ **Open API and a widget** for owners; integration with booking systems.
- ⑤ **Second city**: city file + OSM extract + city layers; prerequisite: open city data (or OSM alone) and a local partner for moderation.

Not another static map – a living, verifiable picture of a city's accessibility.

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